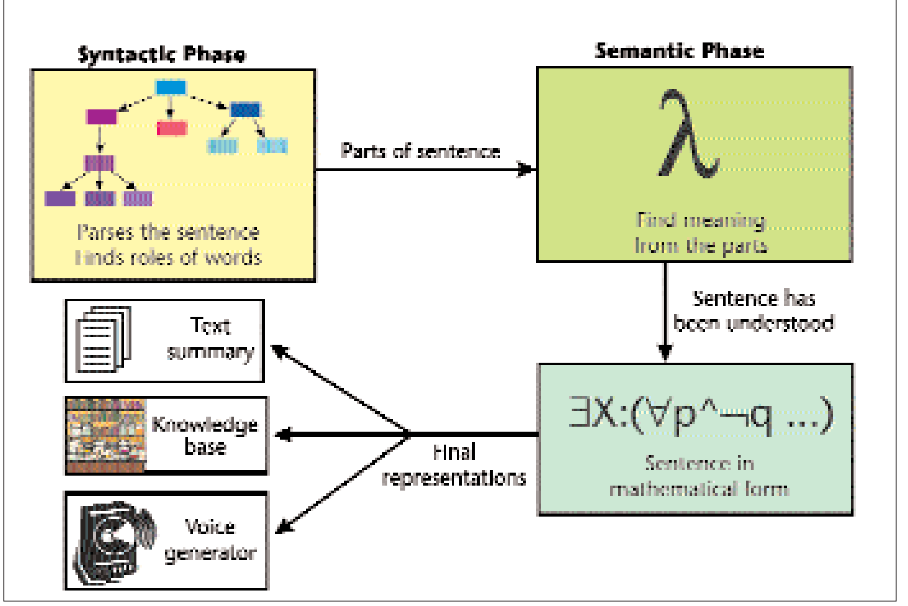
Stages in NLP: The rule-based approach



way of expressing the verb phrase 'likes Garfield' as a predicate—which can then be used in the FOPC representation. Think of a predicate as an incomplete or tentative statement which, when applied to a particular subject, is either true or false. The way to do this could be (lx (x likes(Garfield)). This can take only one argument, namely, 'the x likes Garfield'. If the argument were 'Jon', it would mean 'Jon likes Garfield'. At this point, we have an intermediate representation.

We now need to structure the intermediate representation into a machine-meaningful and logical form. At this point, one might say that the sentence has been, in a sense, understood. The logical form used could be the FOPC. For example, 'Garfield hates all his mice', would be expressed thus:

(∃: M) AND (M: Garfield hates M), where M is a set of mice that belong to Garfield.

The final representation is derived from the semantic representation, and is application specific, as the knowledge gathered usually has to be incorporated into some kind of database.

There are many other things to be taken care of, and the above is only a rough outline. For example, world-knowledge plays an important role, because if you want to understand something someone says, you need to know something about the world we live in. Several AI projects have had as their aim the construction of a common-sense engine. A good example is the cyc project. For more details visit *www.cyc.com*.

The Perfect NLU system

Imagine that you work for a magazine and have perfected NLU systems all around your office. These NLUs would be able to understand each of the italicised words in the snippet given below:

You walk into the office building at 10 PM and speak to the voice recogniser: "Open the gates, and if my friend doesn't get *here* within 3 minutes, close *them*. Oh wait, make *that* 10 minutes, OK?"

You walk in and tell your NLP personal assistant, "Get me all *back* issues that have the phrase 'natural language' in *them*. Oh wait, *forget* the January issue, I don't need *that one*." Your machine can do the correct search, and there you have your back issues.

Now you say, "*We* want to reprint the article which—*now wait*, which one was *that*… yeah, the *one* with references to Garfield. Just *take off* all the images and put *it* on the server."

Next you tell the e-mail server, "Hey, mail everyone *in Digit* tomorrow there's going to be a *reprint*, and tell *them* to do whatever's required."

Come to think of it, if we do get to the point where we can build the ideal NLP, why come into the office at all—you could just do it all over the phone. But hey, if the technology does get this far, how many of us would be needed at all?

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